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ABOUT

AQUATWIN

Real-time Water Intelligence

A low-cost IoT telemetry pipeline that converts invisible aquifer depletion and water quality signals into a live 3D spatial twin for awareness, planning and policy action.

AQUATWIN
SMART WATER FUTURES

IOT TELEMETRY
REAL-TIME INSIGHTS

DEPTH
370.0 cm

DEPLETION RATE
0.025 cm/s

TDS
450.2 ppm

pH
7.63

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PROBLEM & SOLUTIONS

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SDG 6 — Clean Water & Sanitation
SDG 11 — Sustainable Cities & Communities

PROBLEMS

- Groundwater Depletion
- Flood Risk
- Poor Urban Planning
- Water Quality Changes
- Reactive Decision-Making
- Drought & Water Scarcity

SOLUTIONS

- Live Aquifer Digital Twin
- Flood & Saturation Intelligence
- Data-Driven City Planning
- Real-Time Water Quality Monitoring
- Early Warning & Predictive Trends

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HOW IT WORKS

HOW IT WORKS

From sensor data to a live spatial digital twin

01 SENSE

IoT Sensors

Collect data from underground water and environment.

02 CONNECT

Pipeline & Cloud

Transmit and store data securely in real-time.

03 ANALYZE

AI & Simulation

Process data, predict trends and run what-if scenarios.

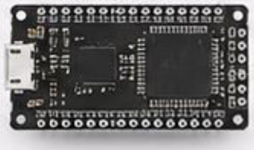
04 VISUALIZE & ACT

3D Twin & Insights

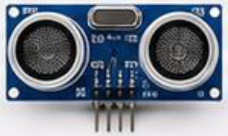
Visualize in 3D, generate alerts and support decision-making.

YOUR HARDWARE SETUP

ESP32-S3



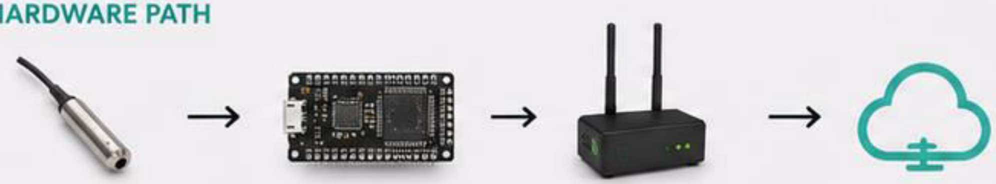
Ultrasonic Sensor



Water Sensor Probe

- Low-power, cost-effective, and reliable
- Real-time underground water monitoring
- Easy to deploy and scale

HARDWARE PATH



Sensor → Controller (ESP32-S3) → Gateway (WiFi/Serial) → Cloud

SOFTWARE PATH

Data Ingestion → Processing & Analytics → Dashboard (Unity 3D) → Insights & Alerts

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EVIDENCE & IMPACT

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WHAT WE PROVED

Real-time sensing and 3D visualization of groundwater conditions.

KEY RESULT

Depth tracking accuracy ±2 cm at 1 Hz streaming rate.

USER BENEFIT

Early detection of depletion trends helps save water & reduce risk.

NEXT MILESTONE

Add rainfall, soil moisture and flood modeling for city-scale twin.

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DEMO & CLOSE

See AquaTwin in action—real data, real insights, real impact.

DEMO

AQUATWIN LIVE DEMO

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THANK YOU

THANK YOU

AquaTwin — Real-time Water Intelligence

Thank you for your time and attention.

Project Name | AquaTwin

Team Name | DATA DRIFTERS

Case Study (Implementation of Groundwater | Indonesia)

https://www.researchgate.net/publication/400495404_Implementation_of_the_Groundwater_Live_Observation_for_Water-Quality_GLOW_in_Bojong_District_Indonesia

Related Research (IOT based Groundwater monitoring)

https://www.researchgate.net/publication/367974113_IoT_based_Ground_Water_Level_Monitoring_System_and_Its_Impact_in_Agricultural_Applications